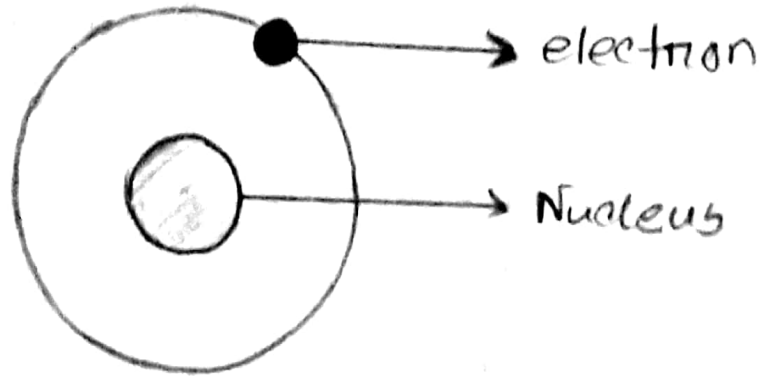


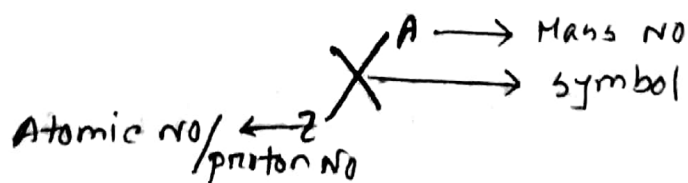
Modern Physics

7 marks

Atom: The smallest unit of matter is called atom.



Atomic Nucleus: The center of an atom is called atomic nucleus. The nucleus consists of proton and neutron. The charge of proton is positive and the charge of neutron is neutral. The value of charge is $1.6 \times 10^{-19} \text{ C}$.



Neutron NO,
 $N = A - Z$

${}^6_6\text{C}^{12} \Rightarrow A = 12$
 $Z = 6$
 $N = 12 - 6 = 6$

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5. Properties of Atomic nucleus:

① Nuclear Mass $= ZM_p + NM_n$

where,

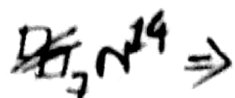
$$M_p = 1.007276 \text{ a.m.u}$$

$$M_n = 1.008665 \text{ a.m.u}$$

$$1 \text{ a.m.u} = 931 \text{ Mev}$$

$$1 \text{ a.m.u} = 1.66054 \times 10^{-27} \text{ kg}$$

$$c^2 = 931 \text{ Mev}$$



$$\text{Nuclear Mass} = ZM_p + NM_n$$

$$= (7 \times 1.007276) + (7 \times 1.008665)$$

$$= 14.11 \text{ a.m.u}$$

$$= 14.11 \times 1.66054 \times 10^{-27}$$

$$= 2.343 \times 10^{-26} \text{ kg.}$$

② Nucleon size :

$$R = r_0 \cdot A^{1/3}$$

where, r_0 = Fermi Constant
 $= 1.3 \times 10^{-15} \text{ m}$

Ex ${}^7_{14}\text{N} \Rightarrow$

$$\begin{aligned}\text{Nucleon size, } R &= r_0 \cdot A^{1/3} \\ &= 1.3 \times 10^{-15} \times 14^{1/3} \\ &= 3.13 \times 10^{-15}\end{aligned}$$

[Note: All size of the universe is 10^{-15}]

③ Nucleon Density:

$$\begin{aligned}\rho_N &= \frac{\text{Nuclear Mass}}{\text{Nuclear Volume}} \\ &= \frac{2M_p + NM_n}{\frac{4}{3}\pi R^3}\end{aligned}$$

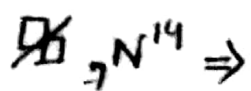
Ex ${}^7_{14}\text{N} \Rightarrow$

$$\begin{aligned}\rho_N &= \frac{2.343 \times 10^{-26}}{\frac{4}{3}\pi \times (3.13 \times 10^{-15})^3} \\ &= 1.824 \times 10^{17} \text{ kgm}^{-3}\end{aligned}$$

[Note: All Density of the universe is 10^{17}]



④ Nuclear charge $= Ze$ $|e| = 1.6 \times 10^{-19} \text{ C}$



Nuclear charge $= Ze$

$$= 7 \times 1.6 \times 10^{-19} \text{ C}$$

$$= 1.12 \times 10^{-18} \text{ C}$$

⑤ Mass Defect:

$$\Delta m = \text{Individual Mass} - \text{Combined Mass}$$

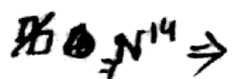
$$= M - A$$

$$\Delta m \text{ of } {}_7\text{N}^{14} = 14.11 - 14$$

$$= 0.11$$

⑥ Binding Energy:

$$B.E = \{ (Zm_p + Nm_n) - A \} \times c^2$$



$$B.E = \{ 14.11 - 14 \} \times 931$$

$$= 102.41 \text{ Mev}$$

⑦ Binding Energy Per nucleon:

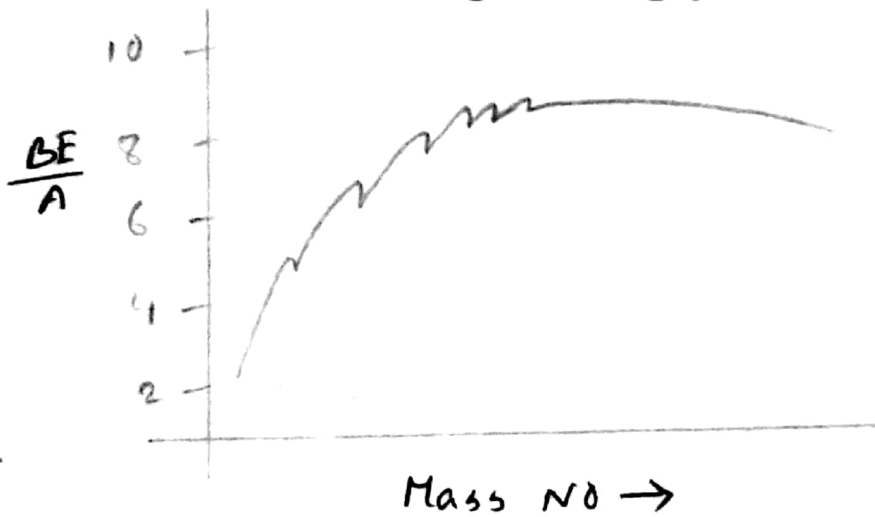
$$= \frac{B.E}{A}$$

✶ ${}_{7}N^{14} \Rightarrow$

$$\frac{B.E}{A} = \frac{102.41}{14}$$

$$= 7.315 \text{ Mev}$$

⑧ Nuclear Binding Energy curve:



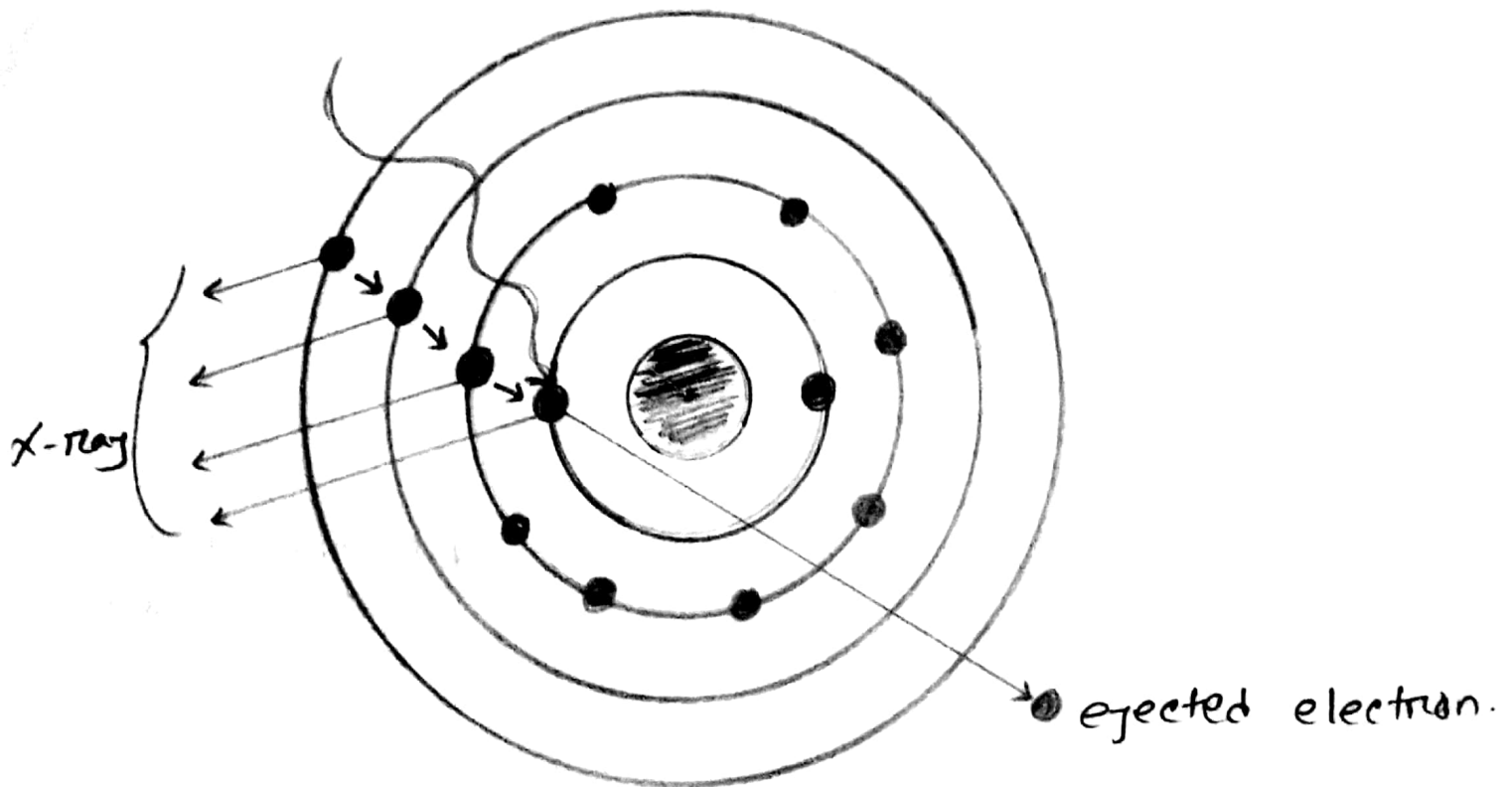
BE তার বেশি
তারপর ৫০ তার বেশি

✶ ${}_{2}He^4$, ${}_{26}Fe^{56}$, ${}_{92}U^{238}$, ${}_{1}H^2$, ${}_{79}Au^{197}$

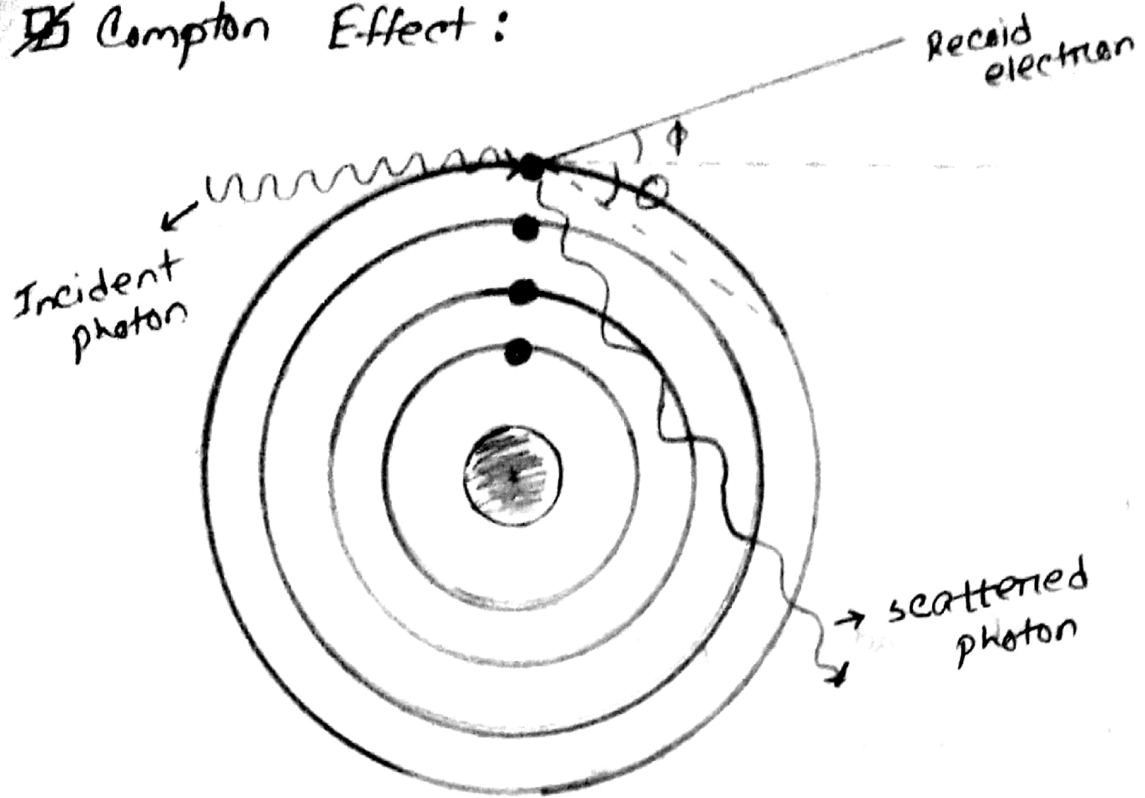
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❖ Photo Electric Effect :

The electron ejected out of metal under the action of light (photon) are known as photo electric effect.



❖ Compton Effect:



❖ X-Ray:

X-Ray is a high energy photon or electromagnetic radiation, having a very short wavelength of the order of 1 a.m.u.

❖ Properties of X-Ray:

- ① Propagated in a straight line with the velocity of 3×10^8
- ② Not deflected by electric or magnetic field.
- ③ Highly penetrating
- ④ Produced fluorescence ~~प्रदीप्ति~~
- ⑤ Destructive effect on living tissues.

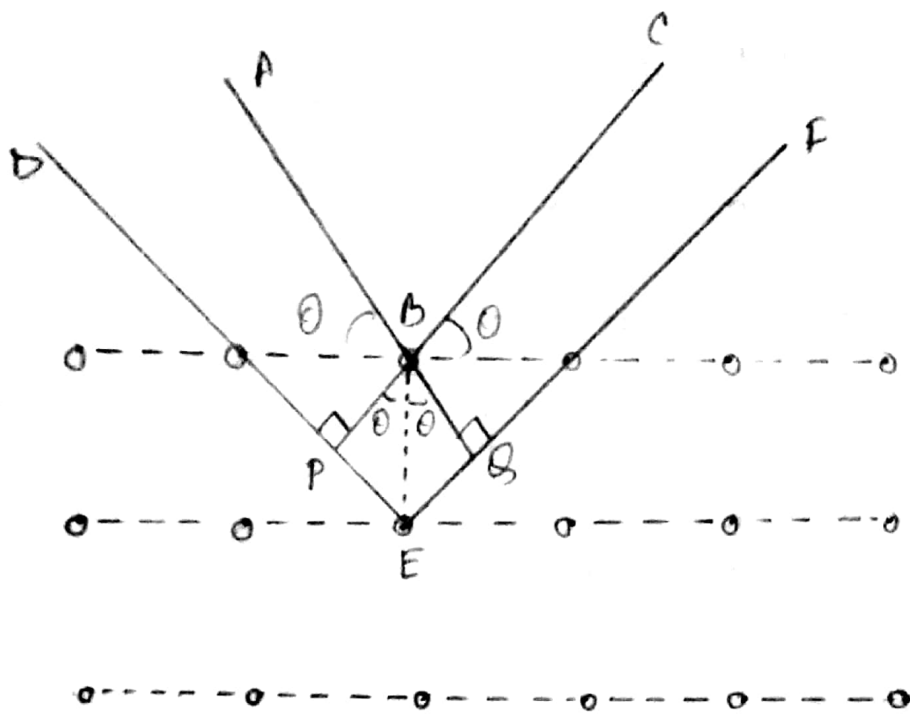


Uses of X-Ray:

- (i) Diagnoses
- (ii) Airport security system.
- (iii) Border security system.
- (iv) X-ray astronomy
- (v) Industrial Radiography
- (vi) X-ray crystallography.

Bragg's Law:

Statement: The intensity of the reflected beam at certain angle will be maximum, where the two reflected wave from two different planes have a path difference equal to an integral multiple of the wavelength of X-ray, while at some other angles, the intensity will be minimum.



path difference = $PE + EQ$

$\triangle PBE$,

$$\sin \theta = \frac{PE}{BE}$$

$$\therefore PE = d \sin \theta$$

$\triangle BQE$,

$$\sin \theta = \frac{EQ}{BE}$$

$$\therefore EQ = d \sin \theta$$

$$\therefore \text{Path difference} = PE + EQ \\ = 2d \sin \theta$$

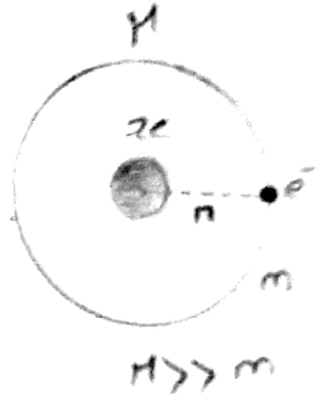
$$\therefore 2d \sin \theta = n\lambda$$

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Postulate of Bohr atom model:
Calculation of the radius atom-

The force of attraction between the nucleus and the electron is,

$$\frac{1}{4\pi\epsilon_0} \cdot \frac{Ze(-e)}{r^2} \quad \text{--- (i)}$$



The ~~centrifugal~~ centrifugal force on the electron = $\frac{mv^2}{r}$ --- (ii)

The system will be stable if,

$$\frac{mv^2}{r^2} = \frac{1}{4\pi\epsilon_0} \cdot \frac{Ze^2}{r^2} \quad \text{--- (iii)}$$

According to Bohr's first postulate,

$$mvr = \frac{nh}{2\pi}$$

$$v = \frac{nh}{2\pi mr}$$

$$v^2 = \frac{n^2 h^2}{4\pi^2 m^2 r^2} \quad \text{--- (iv)}$$

Substituting the value of v in (iii),

$$\frac{m}{r} \left(\frac{n^2 h^2}{4\pi^2 m^2 r^2} \right) = \frac{1}{4\pi\epsilon_0} \cdot \frac{Ze^2}{r^2}$$

$$r = \frac{n^2 h^2}{\pi Ze^2 m} \quad \text{--- (v)}$$

From eqn (2) we find that,

$$r_n \propto n^2$$

The radius of the first orbit for H atom,

$$r_1 = \frac{1^2 \times (6.625 \times 10^{-34})^2 \times (8.854 \times 10^{-12})}{\pi \times (1.6 \times 10^{-19})^2 \times (9.11 \times 10^{-31})} \text{ m}$$

$$= 5.303 \times 10^{-11}$$

$$= 0.5303 \text{ nm}$$

4] Calculation of Total Energy

$$\text{Potential Energy of the } e^- = \frac{-Ze^2}{4\pi\epsilon_0 r}$$

$$\text{Kinetic Energy of the } e^- = \frac{1}{2}mv^2 = \frac{Ze^2}{8\pi\epsilon_0 r}$$

$$\therefore \text{Total Energy} = \frac{-Ze^2}{4\pi\epsilon_0 r} + \frac{Ze^2}{8\pi\epsilon_0 r}$$

$$= \frac{-Ze^2 + Ze^2}{8\pi\epsilon_0 r}$$

$$= \frac{-Ze^2}{8\pi\epsilon_0 r}$$

$$= \frac{-Ze^2}{8\pi\epsilon_0 \cdot \frac{n^2 h^2 \epsilon_0}{\pi \cdot Ze^2 m}}$$

$$\therefore E_n = \frac{-me^4 Z^2}{8\epsilon_0^2 n^2 h^2}$$

Radio Activity

Radioactive elements $\rightarrow$ 83 and above.

Radioactive rays / particle:

i) α - particle (${}_2\text{He}^4$)

ii) β - particle

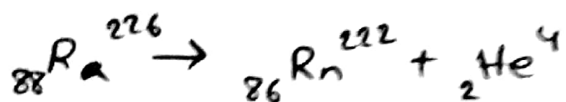
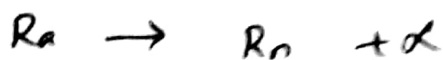
iii) μ - particle

Radioactivity: The spontaneous emission of powerful radiation exhibited by heavy elements is known as radioactivity.

Properties of Radioactive Radiation:

i) Highly penetrating

ii) Irreversible process.

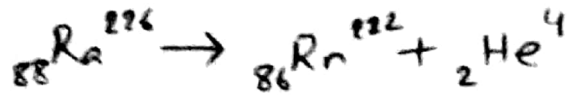
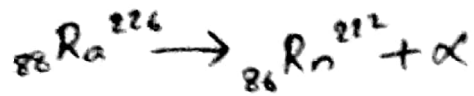


iii) Radiation is spontaneous.

iv) ~~Emission~~ Emission is not instantaneous.

v) There is nothing abnormal of physical and chemical properties.

Q Radioactive Disintegration / decay :



Q Laws of Radioactive Bridge Integration:

2nd law: The rate of Radioactive bridge integration therefore is the number of bridge integration per second is not effected by environmental factors (like temperature, pressure and chemical combination etc) but depends on the number of atoms of the original kind present at any time.

We know,

N_0 = at the begining

N = at any time (t)

dN = at time dt

∴ The rate of disintegration,

$$\frac{dN}{dt} \propto -N$$

$$\therefore \frac{dN}{dt} = -\lambda N$$

$$\text{or, } \frac{dN}{N} = -\lambda dt$$



$$\int \frac{dN}{N} = -\lambda \int dt$$

$$\ln N = -\lambda t + c \quad \text{--- (1)}$$

When $t = 0$, then $N = N_0$

putting value at equation (1),

$$\ln N_0 = c$$

Again putting value of 'c' at equation (1),

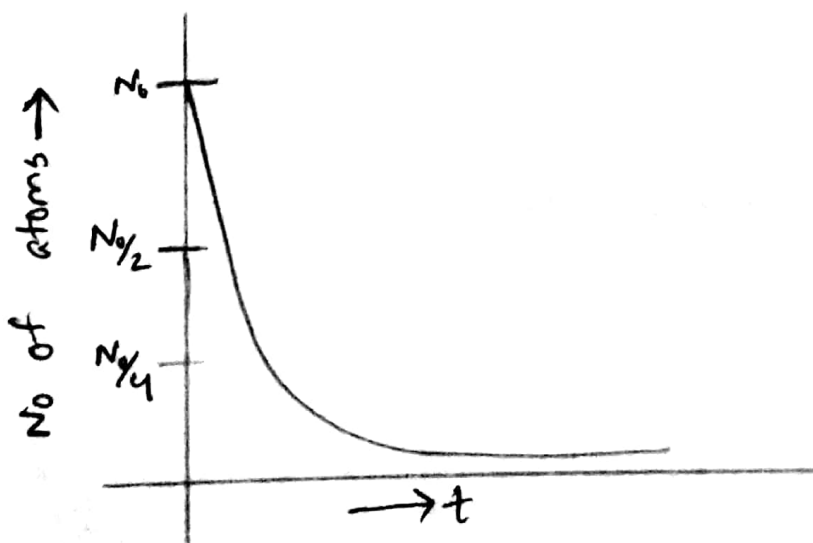
$$\ln N = -\lambda t + \ln N_0$$

$$\ln N - \ln N_0 = -\lambda t$$

$$\ln \frac{N}{N_0} = -\lambda t$$

$$\frac{N}{N_0} = e^{-\lambda t}$$

$$\therefore N = N_0 \cdot e^{-\lambda t}$$



□ Show that the number of radioactive atoms decreases exponentially with time.

☐ Half Life :

The half of a radioactive substance is defined as the time required for $1/2$ of the radioactive substance to disintegration.

From the ~~radio~~ laws of radioactive bridge integration we know,

$$N = N_0 \cdot e^{-\lambda t} \quad \text{--- ①}$$

When $N = \frac{N_0}{2}$ then $t = \frac{1}{2} T_{1/2}$

putting value of N and t at equation ①,

$$\frac{N_0}{2} = N_0 \cdot e^{-\lambda \cdot T_{1/2}}$$

$$\frac{1}{2} = e^{-\lambda \cdot T_{1/2}}$$

$$\ln(1/2) = -\lambda \cdot T_{1/2}$$

$$\therefore T_{1/2} = \frac{0.693}{\lambda}$$

☐ Show that $T_{1/2}$ is inversely proportional to the radio active constant.



Mean Life:

$$\text{Mean Life} = \frac{\text{sum of the lives of all the atoms}}{\text{Total number of atoms}}$$

$$\therefore \text{Total life of } dN \text{ atoms} = (dN)dt \\ = (dN)t$$

$$\text{Total life time of all } N_0 \text{ atoms} = \int_0^{\infty} t \cdot dN$$

$$\therefore \text{Mean Life} = \frac{\text{sum of the lives of all the atoms}}{\text{Total number of atoms}} \\ = \frac{\int_0^{\infty} t \cdot dN}{N_0}$$

We know,

$$N = N_0 \cdot e^{-\lambda t}$$

$$\frac{dN}{dt} = -\lambda N_0 e^{-\lambda t}$$

$$\therefore \cancel{dN} \therefore dN = \lambda N_0 e^{-\lambda t} \cdot dt \quad [\text{ignoring minus}]$$

$$\therefore \text{Mean life, } \bar{T} = \frac{\int_0^{\infty} t \cdot dN}{N_0} \\ = \frac{\int_0^{\infty} t \cdot \lambda N_0 e^{-\lambda t} dt}{N_0} \\ = \lambda \int_0^{\infty} t \cdot e^{-\lambda t} dt$$

$$= \lambda \left[t \int_0^{\infty} e^{-\lambda t} dt - \int_0^{\infty} (t' \int_0^{\infty} e^{-\lambda t} dt) \right]$$

$$= \lambda \left[\left[t \times \frac{e^{-\lambda t}}{-\lambda} \right]_0^{\infty} - \int_0^{\infty} \left\{ \frac{d}{dt} \cdot t \cdot \int_0^{\infty} e^{-\lambda t} dt \right\} dt \right]$$

$$= \lambda \left[\left[\frac{t \cdot e^{-\lambda t}}{-\lambda} \right]_0^{\infty} - \int_0^{\infty} \frac{e^{-\lambda t}}{-\lambda} dt \right]$$

$$= \lambda \left[\left[\frac{t \cdot e^{-\lambda t}}{-\lambda} \right]_0^{\infty} + \frac{1}{\lambda} \int_0^{\infty} \left[\frac{e^{-\lambda t}}{-\lambda} \right]_0^{\infty} \right]$$

$$= \lambda \left[0 - \frac{1}{\lambda^2} [e^{-\infty} - e^0] \right]$$

$$= \lambda \cdot \frac{1}{\lambda^2}$$

$$= \frac{1}{\lambda}$$

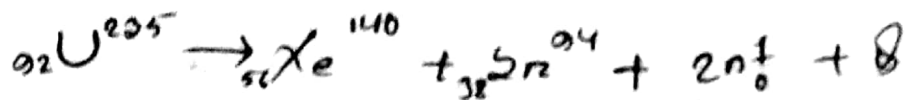
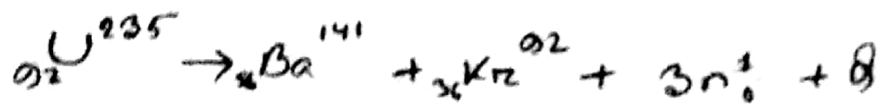
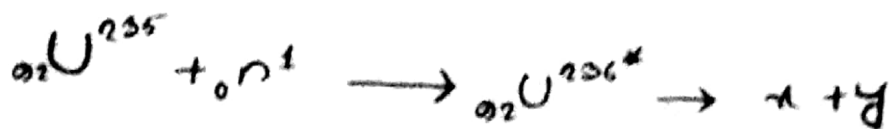
$$\therefore \bar{T} = \frac{1}{\lambda}$$

$$\text{on } \bar{T} = \frac{T_{1/2}}{0.693} = 1.44 \times T_{1/2} \quad [\because T_{1/2} = \frac{0.693}{\lambda}]$$

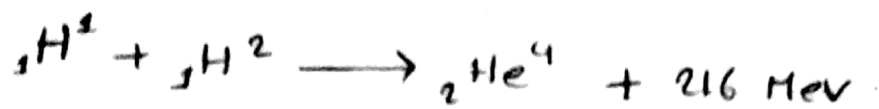
$$\int uv = u \int v dx - \int (u' \int v dx) dx$$



Fission :



Nuclear Fusion :



Nuclear Crosssection:

Q 1 gram radium is reduced by 2.1 mg in 5 years by α decay. Calculate the half-life period of radium?

⇒ We know,

$$N = N_0 \cdot e^{-\lambda t}$$

$$(1 - 2.1 \times 10^{-3}) = 1 \times e^{-\lambda \times 5}$$

$$0.9979 = e^{-5\lambda}$$

$$\ln(0.9979) = -5\lambda$$

$$-2.1022 \times 10^{-3} = -5\lambda$$

$$\therefore \lambda = 4.204 \times 10^{-4}$$

Again,

$$T_{1/2} = \frac{0.693}{\lambda} = \frac{0.693}{4.204 \times 10^{-4}} = 1648.48 \text{ (Ans)}$$

☐ The half-life period of radium is 1500 years. In how many years will 1 gram of pure element

a) lose 1 centigram

b) be reduced to 1 centigram.

(a)

$$T_{1/2} = \frac{0.693}{\lambda}$$

$$\lambda = \frac{0.693}{T_{1/2}} = \frac{0.693}{1500} = 4.36 \times 10^{-4} \text{ y}^{-1}$$

We know,

$$N = N_0 e^{-\lambda t}$$

$$0.01 N_0 = N_0 \cdot e^{-\lambda t}$$

$$\therefore e^{-\lambda t} = 0.01$$

$$-\lambda t = \ln(0.01)$$

$$t = -\frac{1}{\lambda} \ln(0.01) = -\frac{1}{4.36 \times 10^{-4}} \ln(0.01)$$

$$= 23.0512 \text{ years.}$$

given,

$$T_{1/2} = 1500 \text{ years}$$

$$N = (1 - 0.01) N_0$$

$$= 0.99 N_0$$

(b)

$$N = N_0 e^{-\lambda t}$$

$$0.01 = e^{-\lambda t}$$

$$\ln(0.01) = -\lambda t$$

$$t = \frac{1}{-\lambda} \ln(0.01)$$

$$= \frac{1}{-4.36 \times 10^{-4}} \times \ln(0.01)$$

$$= 10562.3 \text{ years.}$$

$$\left\{ \begin{array}{l} N = 0.01 \text{ gm} \\ N_0 = 1 \\ \frac{N}{N_0} = 0.01 \end{array} \right.$$

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✓ Calculate the time requirement required for 10% of a sample of thorium to disintegrate. Assume the half life of thorium to be 1.4×10^{10} years (2.12890529×10^9 year).

⇒ given that,

$$T_{1/2} = 1.4 \times 10^{10}$$

$$T_{1/2} = \frac{0.693}{\lambda}$$

$$\therefore \lambda = \frac{0.693}{1.4 \times 10^{10}}$$

$$= 4.95 \times 10^{-11} \text{ s}^{-1}$$

$$\frac{N}{N_0} = (100 - 10)\% = 90\%$$

$$\therefore \frac{N}{N_0} = \frac{90}{100} = 0.9$$

∴

$$N = N_0 \cdot e^{-\lambda t}$$

$$\frac{N}{N_0} = e^{-\lambda t}$$

$$0.9 = e^{-\lambda t}$$

$$\ln(0.9) = -\lambda t$$

$$\therefore t = -\frac{1}{\lambda} \ln(0.9)$$

$$= -\frac{1}{4.95 \times 10^{-11}} \ln(0.9)$$

$$= 2.128 \times 10^9 \text{ year.}$$

Solid State Physics

10 Marks

Q1 Crystalline Solid:

In crystalline solids, the atoms are stacked in a regular manner in crystals.

Q2 Polycrystalline Solid:

Polycrystalline materials are solids that consist of many small crystals (the grains).

Q3 Non-crystalline Solid:

In ~~non-crystalline~~ non-crystals the same atomic groups are arranged more randomly.

Q4 Single crystal:

In single crystals the periodicity extends throughout the material.

Q5 Lattice:

The regular periodic arrangements of atoms, ions and molecules in space which looks like a net is called lattice.

Q6 Lattice Point:

If each atom in a lattice is replaced by a point then such a point is called a lattice point. That is, every point in a regular periodic array in space is called a lattice point.



Basis:

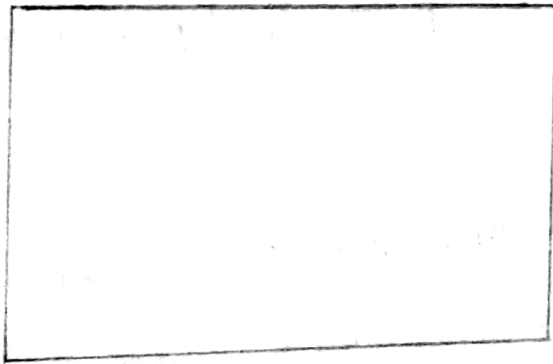
The structure of all crystal can be described in terms of a lattice with group of atoms attached to every lattice point. The group of atoms is called basis.

Crystal:

A crystal is a regular array of three dimensional units. The structural units are atoms, ions, and molecules or group of atoms.

Unit Cell:

Crystal System:



⇒ Right down properties of crystal system
ସମସ୍ତ Exam 27
କ୍ଷେତ୍ର crystal system
no of crystal system
ଆଉ example
ମିଆଳ.

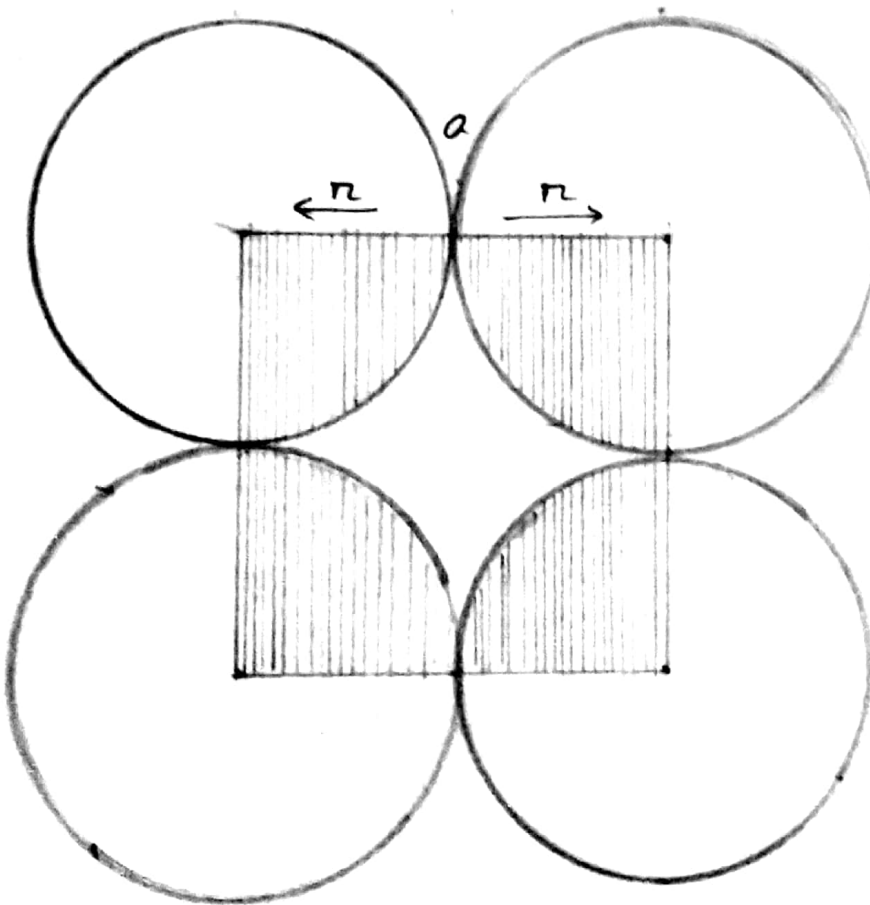
Packing Fraction:

Packing fraction is defined as the ratio of the volume of the atoms per unit cell to the total volume of the unit cell.

$$P.F. = \frac{\text{The volume of the atoms per unit cell}}{\text{The total volume of the unit cell.}}$$

i) Simple Cubic:

(Imp)



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If a is the side of the cell and r is the radius of the atom then from figure,

$$a = 2r$$

Then,

Total volume of the unit cell is,

$$a^3 = (2r)^3 = 8r^3$$

The number of atoms per unit cell,

$$\frac{1}{8} \times 8 = 1$$

$\therefore$ The volume of the atoms per unit cell,

$$\frac{4}{3} \times \pi r^3$$

$$\therefore \text{Packing fraction} = \frac{\text{The volume of the atoms per unit cell}}{\text{The total volume of the unit cell.}}$$

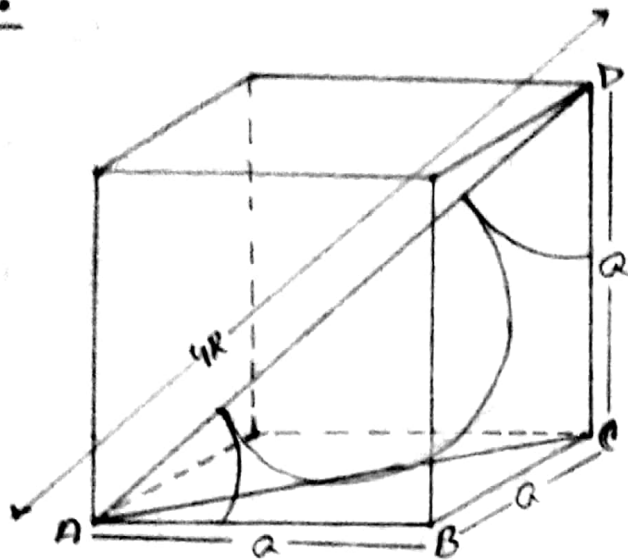
$$= \frac{\frac{4}{3} \pi r^3}{8r^3}$$

$$= \frac{\pi}{6}$$

$$= 0.53$$

11) Bee:

(Imp)



From figure, it can be written as,

$$AC^2 = a^2 + a^2$$

$$AC = a\sqrt{2}$$

$$AD = r + 2r + r \\ = 4r$$

Again,

$$AD^2 = AC^2 + CD^2$$

$$(4r)^2 = 2a^2 + a^2$$

$$16r^2 = 3a^2$$

$$a^2 = \frac{16}{3}r^2$$

$$a = \frac{4}{\sqrt{3}}r$$

Then, total volume of the unit cell is,

$$a^3 = \left(\frac{4}{\sqrt{3}}r\right)^3$$

$$= \frac{64}{3\sqrt{3}}r^3$$

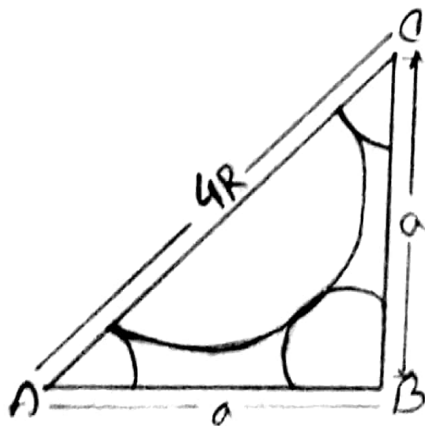
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The number of atoms per unit cell = 2

The volume of atoms per unit cell = $2 \times \frac{4}{3} \pi r^3$
 $= \frac{8}{3} \pi r^3$

$$\therefore \text{Packing fraction} = \frac{\frac{8}{3} \pi r^3}{\frac{64}{3\sqrt{3}} r^3}$$
$$= \frac{\pi\sqrt{3}}{8}$$
$$= 0.68.$$

iii) fcc:
(5m)



from figure,

$$AC = r + 2r + r = 4r$$

$$AC^2 = a^2 + a^2 = 2a^2$$

$$\therefore (4r)^2 = 2a^2$$

$$\therefore a^2 = \frac{16r^2}{2}$$

$$a^2 = 8r^2$$

$$a = 2\sqrt{2}r$$

Then,

total volume of the unit cell is,

$$a^3 = (2\sqrt{2}\pi)^3$$
$$= 16\sqrt{2}\pi^3$$

Then,

The number of atoms per unit cell = 4

The volume of atoms per unit cell = $4 \times \frac{4}{3} \pi r^3$

$$= \frac{16}{3} \pi r^3$$

$$\therefore \text{Packing fraction} = \frac{\frac{16}{3} \pi r^3}{\frac{16\sqrt{2}}{1} r^3}$$
$$= \frac{1}{3\sqrt{2}} \pi$$
$$= 0.74$$

Coordination Number:

Every atom in a crystalline structure is surrounded by other atoms. The number of nearest neighbours that surround each atom is called coordination number.

at NaCl,

$$\text{Na: } 000, \frac{1}{2} \frac{1}{2} 0, \frac{1}{2} 0 \frac{1}{2}, 0 \frac{1}{2} \frac{1}{2}$$

$$\text{Cl: } \frac{1}{2} \frac{1}{2} \frac{1}{2}, 00 \frac{1}{2}, 0 \frac{1}{2} 0, \frac{1}{2} 00$$



CCl₄ Structure:

From PDF

Miller indices:

In order to represent any particular crystal plane in terms of the crystallographic axes and the intercepts of the plane on these axes, the reciprocal of the intercepts are taken on the three axes in terms of the lattice constants and these reciprocals are reduced to the smallest three integers having the same ratio. These integers are the Miller indices of the plane.

Procedure for finding miller indices:

Example:

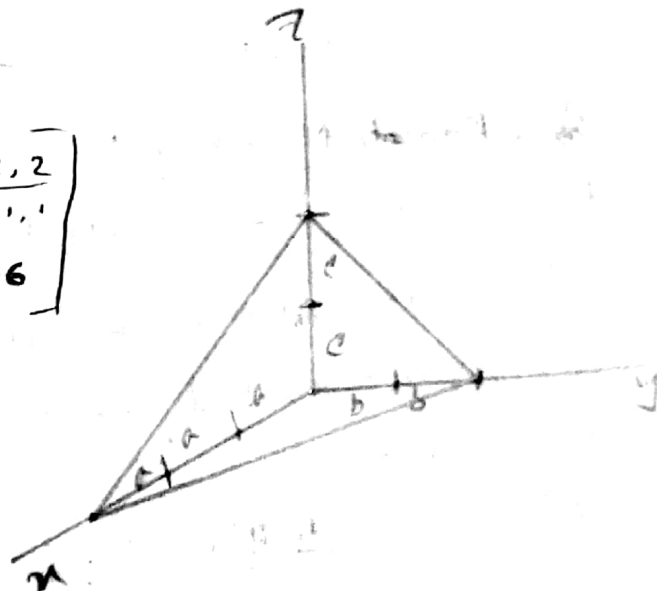
$$3a, 2b, 2c$$

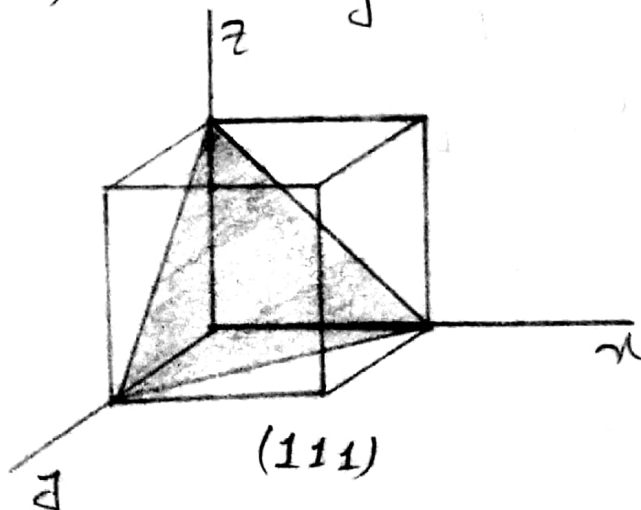
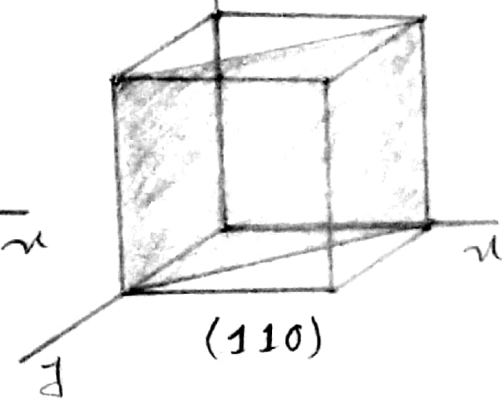
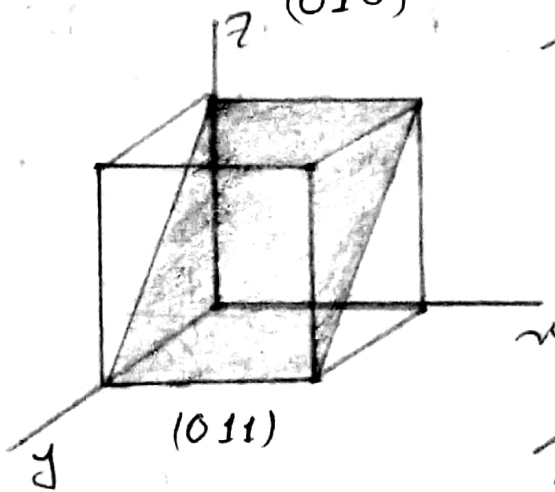
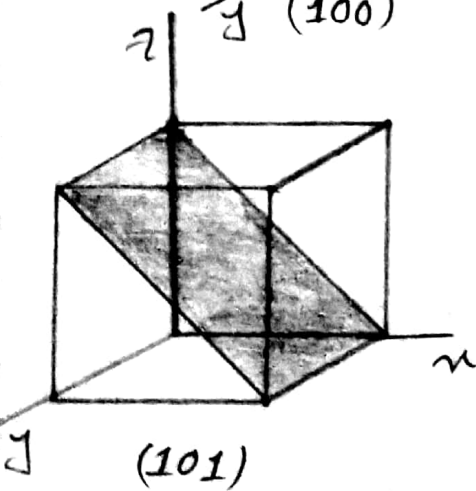
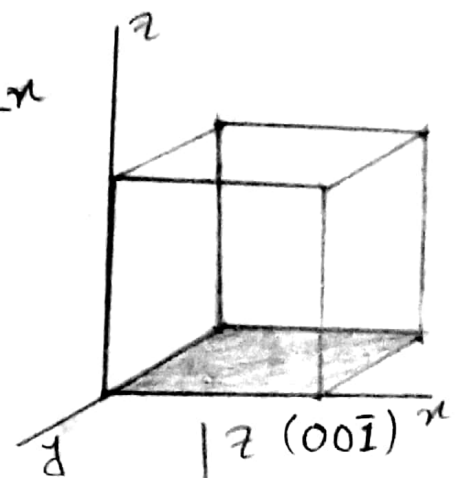
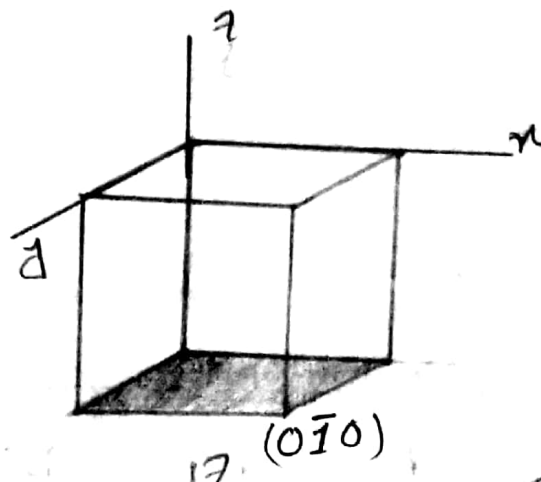
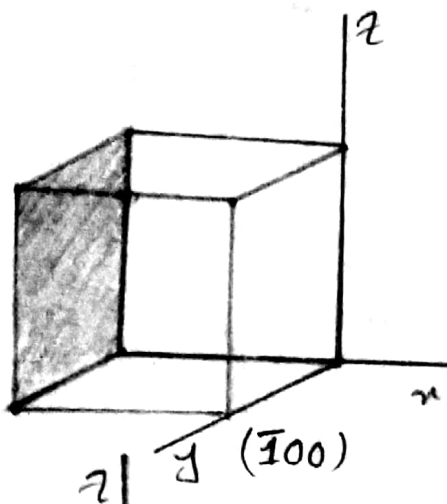
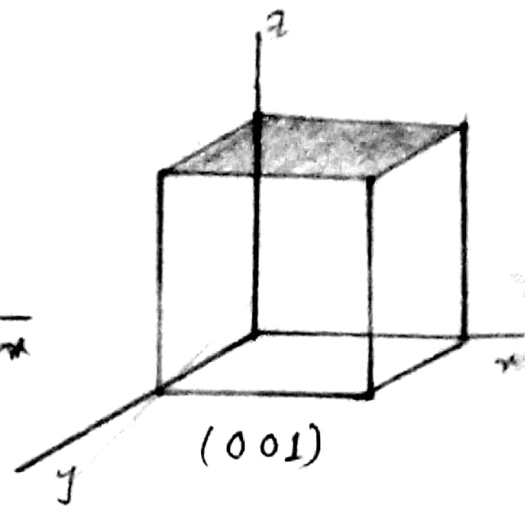
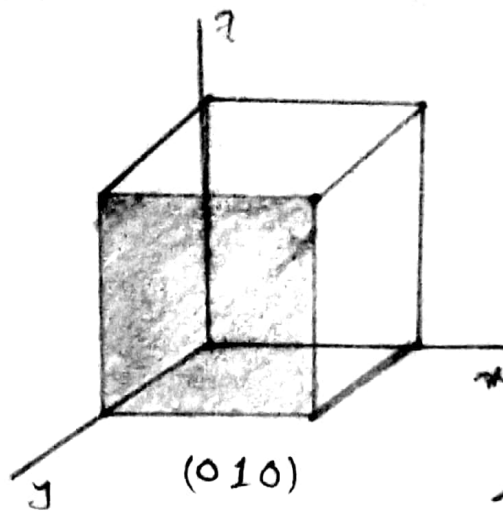
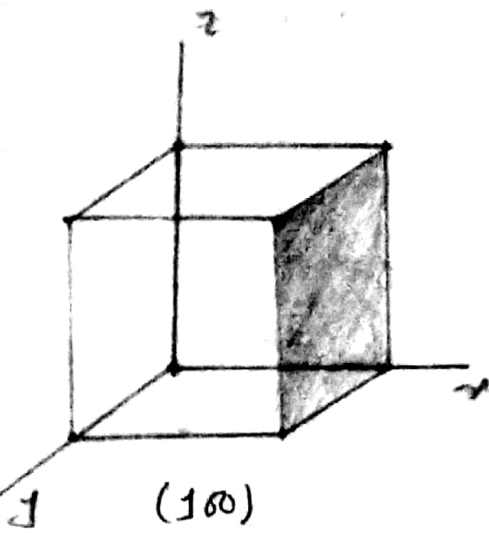
$$\Rightarrow \frac{1}{3}, \frac{1}{2}, \frac{1}{2}$$

$$\Rightarrow \frac{1}{3} \times 6, \frac{1}{2} \times 6, \frac{1}{2} \times 6$$

$$\Rightarrow (233)$$

$$\left[\begin{array}{c} 23, 2, 2 \\ 3, 1, 1 \\ 2 \times 3 = 6 \end{array} \right]$$

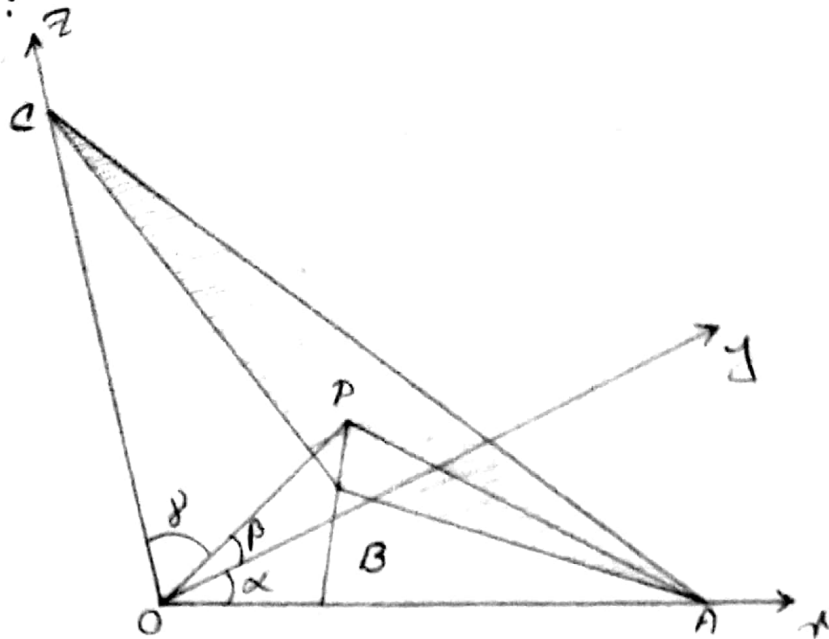




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Relation between Inter-planar spacing and miller indices:

Exam 4 PDF of
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The intercepts of the plane on the three axes are,
 $OA = \frac{a}{h}$, $OB = \frac{b}{k}$ and $OC = \frac{c}{l}$.

$$\cos \alpha = \frac{d}{OA} = \frac{d}{a/h} = \frac{dh}{a}$$

$$\cos \beta = \frac{d}{OB} = \frac{d}{b/k} = \frac{dk}{b}$$

$$\cos \gamma = \frac{d}{OC} = \frac{d}{c/l} = \frac{dl}{c}$$

$$\therefore \cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1$$

$$\left(\frac{dh}{a}\right)^2 + \left(\frac{dk}{b}\right)^2 + \left(\frac{dl}{c}\right)^2 = 1$$

For cubic Lattice, $a = b = c$,

$$\therefore d = \frac{a}{\sqrt{h^2 + k^2 + l^2}}$$

Relativity

Inertial Frame:

The frames with respect to which an unaccelerated body appears unaccelerated are inertial frames. In other words, the frames which are at rest or in uniform translatory motion relative to one another are inertial frames. In brief, we may say "An inertial frame is one in which Newton's first law is true."

Non-inertial Frame:

The frames with respect to which an unaccelerated body appears ~~accelerate~~ accelerated are called non-inertial frames. In other words, the accelerated frames are non-inertial.

Postulates of Special theory of Relativity:

The special theory of relativity is based on the two postulates.

Postulate-I:

The fundamental laws of physics have the same form for all inertial system, for all reference systems at rest or moving with constant linear velocity relative to one another.

Postulate-II:

The velocity of light in free space has the same value for all inertial observers.



❑ Lorentz Transformation Equations:

$$x' = \frac{x - vt}{\sqrt{1 - v^2/c^2}}$$

$$y' = y$$

$$z' = z$$

$$t' = \frac{t - vx/c^2}{\sqrt{1 - v^2/c^2}}$$

❑ Inverse Lorentz Transformation Equation:

$$x = \frac{x' + vt'}{\sqrt{1 - v^2/c^2}}$$

$$y = y'$$

$$z = z'$$

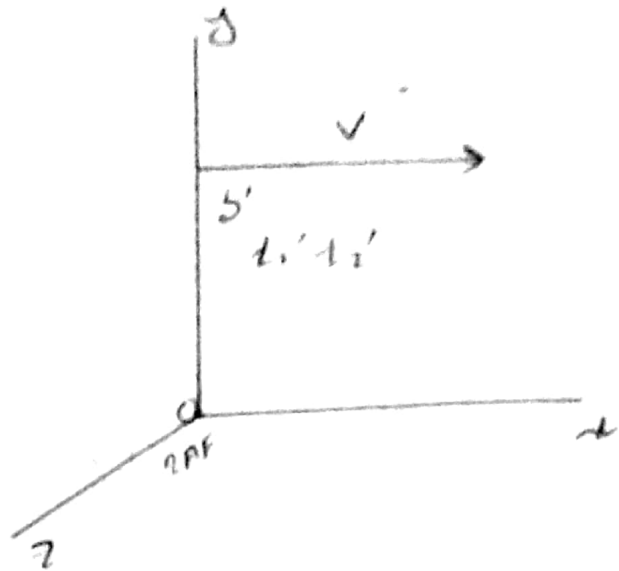
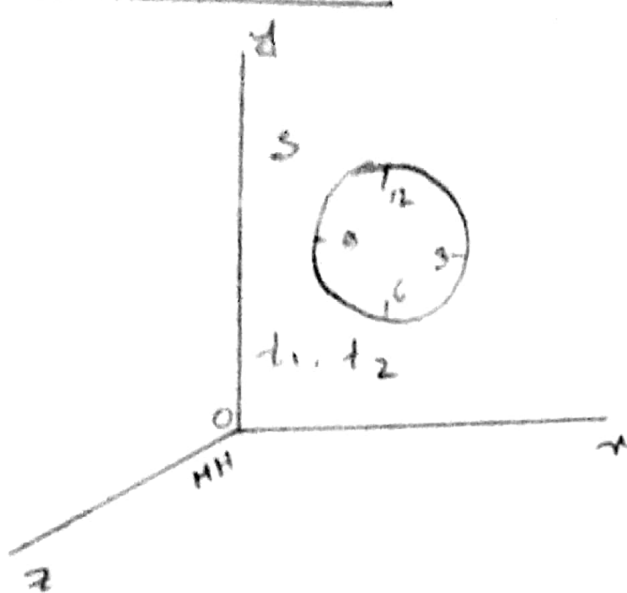
$$t = \frac{t' + vx'/c^2}{\sqrt{1 - v^2/c^2}}$$

❑

$$m = \frac{m_0}{\sqrt{1 - v^2/c^2}}$$



Time Dilation:



Let a clock be placed in reference system S .
Corresponding to the time t_1 ,

$$t_1' = \frac{t_1 - \frac{vx_1}{c^2}}{\sqrt{1 - \frac{v^2}{c^2}}} \quad \text{--- (1)}$$

If this clock Again gives a signal at time t_2 , then corresponding time t_2' in frame S' is given by,

$$t_2' = \frac{t_2 - \frac{vx_2}{c^2}}{\sqrt{1 - \frac{v^2}{c^2}}} \quad \text{--- (2)}$$

time interval between t_1' and t_2' ,

$$t_2' - t_1' = \frac{t_2 - \frac{vx}{c^2}}{\sqrt{1 - \frac{v^2}{c^2}}} - \frac{t_1 - \frac{vx}{c^2}}{\sqrt{1 - \frac{v^2}{c^2}}}$$

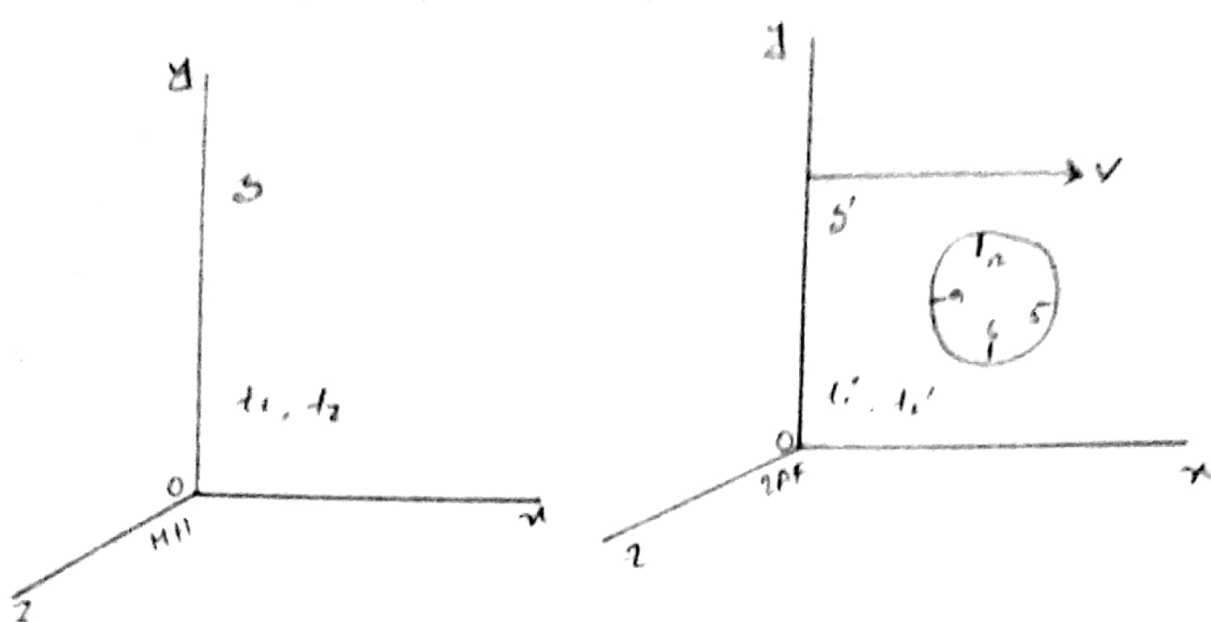
$$= \frac{t_2 - \frac{vx}{c^2} - t_1 + \frac{vx}{c^2}}{\sqrt{1 - \frac{v^2}{c^2}}}$$

$$= \frac{t_2 - t_1}{\sqrt{1 - \frac{v^2}{c^2}}}$$

$$\therefore \Delta t' = \frac{\Delta t}{\sqrt{1 - \frac{v^2}{c^2}}}$$

$$\Delta t' (\sqrt{1 - \frac{v^2}{c^2}}) = \Delta t$$

$$\therefore \Delta t' > \Delta t$$



Let a clock be in reference system S' . Corresponding to the time according to inverse Lorentz Transformation equation,

$$t_1 = \frac{t_1' + \frac{vx_1'}{c^2}}{\sqrt{1 - \frac{v^2}{c^2}}} \quad \text{--- (1)}$$

$$t_2 = \frac{t_2' + \frac{vx_2'}{c^2}}{\sqrt{1 - \frac{v^2}{c^2}}} \quad \text{--- (2)}$$

Object at
respect to
observer (-)ve
2(m) inverse

time interval between t_1 and t_2 ,

$$\begin{aligned} t_2 - t_1 &= \frac{t_2' + \frac{vx_2'}{c^2}}{\sqrt{1 - \frac{v^2}{c^2}}} - \frac{t_1' + \frac{vx_1'}{c^2}}{\sqrt{1 - \frac{v^2}{c^2}}} \\ &= \frac{t_2' - t_1'}{\sqrt{1 - \frac{v^2}{c^2}}} \end{aligned}$$

$$\therefore \Delta t = \frac{\Delta t'}{\sqrt{1 - \frac{v^2}{c^2}}}$$

$$\Delta t \left(\sqrt{1 - \frac{v^2}{c^2}} \right) = \Delta t'$$

$$\Delta t > \Delta t'$$

Finally we can say a moving clock always appears to be ~~slow~~ slow.

Length Contraction:



$$L_0 = x_2 - x_1 \quad \text{--- (1)}$$

$$L = x_2' - x_1' \quad \text{--- (2)}$$

$$x_1 = \frac{x_1' + vt^2}{\sqrt{1 - \frac{v^2}{c^2}}} \quad \text{--- (3)}$$

$$x_2 = \frac{x_2' + vt^2}{\sqrt{1 - \frac{v^2}{c^2}}} \quad \text{--- (4)}$$

$$\begin{aligned} x_2 - x_1 &= \frac{x_2' + vt^2}{\sqrt{1 - \frac{v^2}{c^2}}} - \frac{x_1' + vt^2}{\sqrt{1 - \frac{v^2}{c^2}}} \\ &= \frac{x_2' - x_1'}{\sqrt{1 - \frac{v^2}{c^2}}} \end{aligned}$$

$$\therefore L_0 = \frac{L}{\sqrt{1 - \frac{v^2}{c^2}}}$$

$$L_0 \left(\sqrt{1 - \frac{v^2}{c^2}} \right) = L$$

$$\therefore L_0 > L$$



□ Einstein's Mass-Energy relation:

From the work-energy theorem,

$$W = \int_0^v \vec{F} \cdot d\vec{s} \quad \text{--- (1)}$$

Kinetic Energy of the body,

$$\begin{aligned} E_k = T = W &= \int_0^v \vec{F} \cdot d\vec{s} \\ &= \int_0^v m a \, ds \\ &= \int_0^v m \frac{dv}{dt} \, ds \\ &= \int_0^v m \cdot dv \cdot \frac{ds}{dt} \\ &= \int_0^v m \, dv \cdot v \\ &= \int_0^v \vec{v} \cdot d(m\vec{v}) \quad \text{--- (2)} \end{aligned}$$

We know,

$$m = \frac{m_0}{\sqrt{1 - \frac{v^2}{c^2}}}$$

$$\begin{aligned} \therefore E_k = T &= \int_0^v \vec{v} \cdot d \left[\frac{m_0}{\sqrt{1 - v^2/c^2}} \cdot \vec{v} \right] \\ &= m_0 \int_0^v \vec{v} \cdot d \left[\vec{v} \frac{1}{\sqrt{1 - v^2/c^2}} \right] \\ &= m_0 \int_0^v \vec{v} \cdot d \left[\vec{v} \left(1 - \frac{v^2}{c^2} \right)^{-1/2} \right] \end{aligned}$$

$$= m_0 \int_0^v \vec{v} \left[\left(1 - \frac{v^2}{c^2}\right)^{-1/2} \cdot dv - v \cdot \frac{1}{2} \left(1 - \frac{v^2}{c^2}\right)^{-3/2} \cdot \left(-\frac{2v}{c^2}\right) dv \right]$$

$$= m_0 \int_0^v \vec{v} \left[\left(1 - \frac{v^2}{c^2}\right)^{-1/2} \cdot dv + \frac{v^2}{c^2} \left(1 - \frac{v^2}{c^2}\right)^{-3/2} \cdot dv \right]$$

$$= m_0 \int_0^v \vec{v} \left(1 - \frac{v^2}{c^2}\right)^{-3/2} \left[1 - \frac{v^2}{c^2} + \frac{v^2}{c^2}\right] dv$$

$$E_k = T = m_0 \int_0^v \frac{\vec{v} dv}{\left(1 - \frac{v^2}{c^2}\right)^{3/2}} \quad \text{--- (3)}$$

Let,

$$y = 1 - \frac{v^2}{c^2}$$

$$\frac{dy}{dv} = \frac{d}{dv} \left(1 - \frac{v^2}{c^2}\right)$$

$$= \frac{-2v}{c^2}$$

$$\therefore v dv = - \frac{c^2 dy}{2}$$

When, $v=0$, then $y=1$

When, $v=v$, then $y = 1 - \frac{v^2}{c^2}$

Therefore from equation-(3),

$$E_k = T = m_0 \int_1^{1 - \frac{v^2}{c^2}} \frac{-c^2 dy/2}{y^{3/2}}$$

$$= m_0 \times \left(\frac{-c^2}{2}\right) \int_1^{1 - \frac{v^2}{c^2}} \frac{dy}{y^{3/2}}$$



$$= \frac{-m_0 c^2}{2} \left[\frac{y^{-\frac{3}{2}+1}}{-\frac{3}{2}+1} \right]_1^{1-\frac{v^2}{c^2}}$$

$$= \frac{-m_0 c^2}{2} \left(-\frac{1}{2}\right) \left[\frac{1}{\sqrt{y}} \right]_1^{1-\frac{v^2}{c^2}}$$

$$= m_0 c^2 \left[\frac{1}{\sqrt{1-\frac{v^2}{c^2}}} - 1^{-1/2} \right]$$

$$= \left[\frac{m_0}{\sqrt{1-\frac{v^2}{c^2}}} - m_0 \right] c^2$$

$$= (m - m_0) c^2; \text{ This is the relativistic expression of kinetic energy.}$$

We know,

$$E = \text{Rest energy} + \text{Relativistic energy}$$

$$= m_0 c^2 + (m - m_0) c^2$$

$$= m_0 c^2 + m c^2 - m_0 c^2$$

$\therefore E = m c^2$; which is the Einstein's mass-energy relation.

Defects Bonds in Solid

The term defect denotes a state of the lattice in which the atomic arrangement has departed from regularity.

1. Point defects
2. Line defects
3. Plane defects
4. Electronic Imperfections
5. Excitation state of crystals
6. Transient imperfection.

① Point Defects:

If the deviation from regularity is confined to

- i) Interstitial atom
- ii) Schottky defect or vacancy
- iii) Frenkel defect
- iv) Substitutional impurity atom

② Line Defects:

Defects in dislocation.

③ Plane Defects:

Have an extension in area.

- i) Lineage boundary
- ii) Grain boundary
- iii) Stacking boundary.



④ Electronic Imperfection:

Errors in charge distribution or energy.

⑤ Excitation States of crystals:

Deviations from perfect crystal symmetry.

- i) Phonons and magnons
- ii) Conduction electron and hole
- iii) Excitations
- iv) Quantised plasma waves.

⑥ Transient Imperfection:

Imperfection are introduced into the crystal from external sources.

☐ Distinguish between Schottky and Frenkel defects:

Schottky Defects

Frenkel Defects

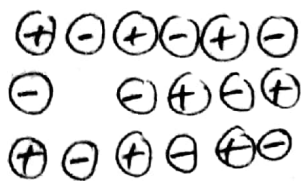
- | | |
|--|--|
| ① This is a defect in which an atom or ion is found to be missing from its correct position. | ① When an atom or ion leaves its normal site and is found to occupy another position in the interstice, we get a Frenkel defect. |
| ② In pure alkali halides the most common lattice vacancies are Schottky defects. | ② In pure silver halides the most common vacancies are Frenkel defects. |
| ③ The production of Schottky defects lowers the density of | ③ The production of Frenkel defects does not change the |

the crystal because the volume is increased with no increasing of mass.

④ After various degrees of dissociation the two components of Schottky defects are not independent.

⑤ There are the individual positive and negative or both types of ion vacancies. Therefore, Schottky defects only vacancy.

Positive ion missing



Negative ion missing



Pair of ions missing

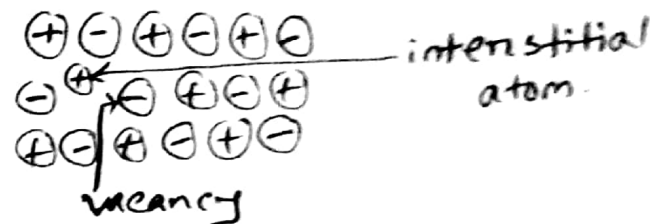


volume of crystal, thus the density remains unchanged.

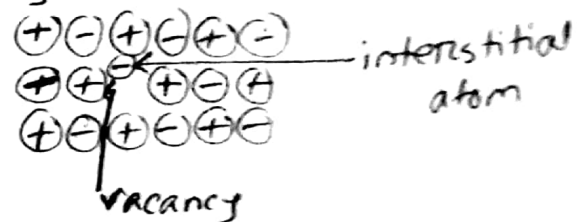
④ After various degrees of dissociation the two components of the Frenkel defects are independent of each other.

⑤ In this case two imperfection is created, an interstitial atom and a vacancy.

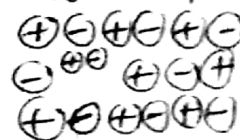
positive ion displacement & vacancy



negative ion displacement & vacancy



Pair of ion displacement & vacancy



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Crystal binding

Crystal binding is formed by the attractive electrostatic interaction between the negative charges of the electrons and the positive charges of the nuclei which is entirely responsible for the ~~at~~ cohesion of solids. Magnetic forces have only a weak effect on cohesion and gravitational forces are negligible.

The strength of crystal bond is measured by the energy required to dissociate a solid into isolated atoms, molecules or ions an appropriate and the energy is generally known as cohesive energy.

Different types of crystal bindings:

① Ionic Bond:

Ionic crystals are made up of positive and negative ions. The ionic bond results from the electrostatic interaction of oppositely charged ions. Example: NaCl , CsCl , etc.

② Co-valent Bond:

The co-valent bond is the classical electron pair or homopolar bond of chemistry, particularly in organic chemistry. The cohesive forces in co-valent crystals arise from the sharing of electrons by adjacent atoms. Each atom that participates in a co-valent bond contributes an electron to

the bond. Example: Diamond, Silicon, Germanium etc.

③ Metallic Bond:

The valance electrons of metal atoms are weakly bound. When such atoms interact to become a crystal, their valence electrons form a 'gas' of electron that move with relative freedom through the resulting assembly of metal ions. The characteristics feature of the metallic binding is the lowering of the energy of the valence electrons in the metal as compared with the free atom.

Example: Cu, Ag, Au, etc.

④ Vander waals Bond:

All atoms and molecules exhibit weak, short-range attractions for one another due to vander waals forces. The bond formed by the vander waals interaction is known as vander waals bond. vander waals interaction are responsible for the condensation of gases into liquids and freezing of liquids into solid in the absence of ionic co-valent or metallic bonding mechanisms.

Example: He, Ne, Ar, etc.



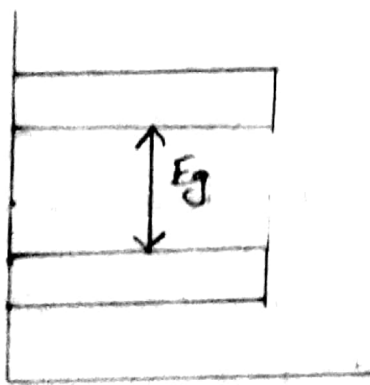
⑤ Hydrogen Bond:

A neutral hydrogen bond has only electron. It should form a co-valent with only one other atom.

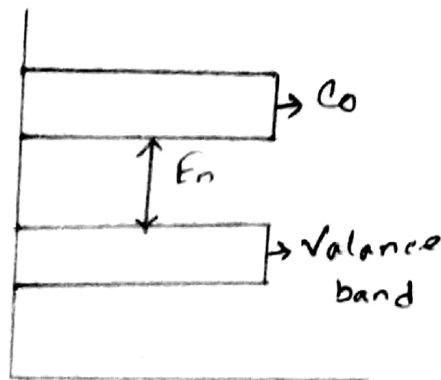
However, under ~~set~~ certain condition an atom of hydrogen is attracted by rather strong forces to two atoms, thus forming a hydrogen bond between them, with a bond energy of the order of 0.1 eV. A hydrogen bond occurs between certain molecules containing hydrogen atoms.

Example: HF_2^+ , H_2O etc.

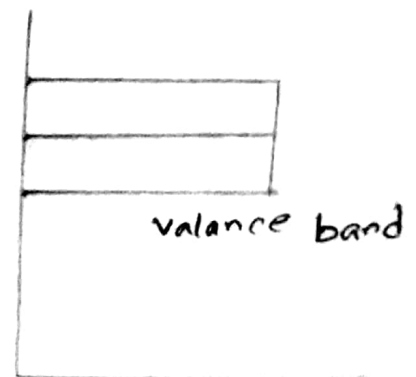
✶ Energy Diagram :



Insulator



Semi conductor



Conductor

☒ Cohesive Energy :

